OrCAD Installation guide.

1. Download the OrCAD software from the link on the module website (ftp://ftp.ee.up.ac.za/pub/windows/eda/OrCAD_Demo_160.zip).

2. Right-click on the downloaded file and select “extract” in order to unzip the file to a known location on your computer.

3. From inside the extracted folder, double click on the “Setup” application file. This will start the installation process.

4. Follow the instructions and select where you would like the software to be installed.

5. Let the software install (this may take some time).

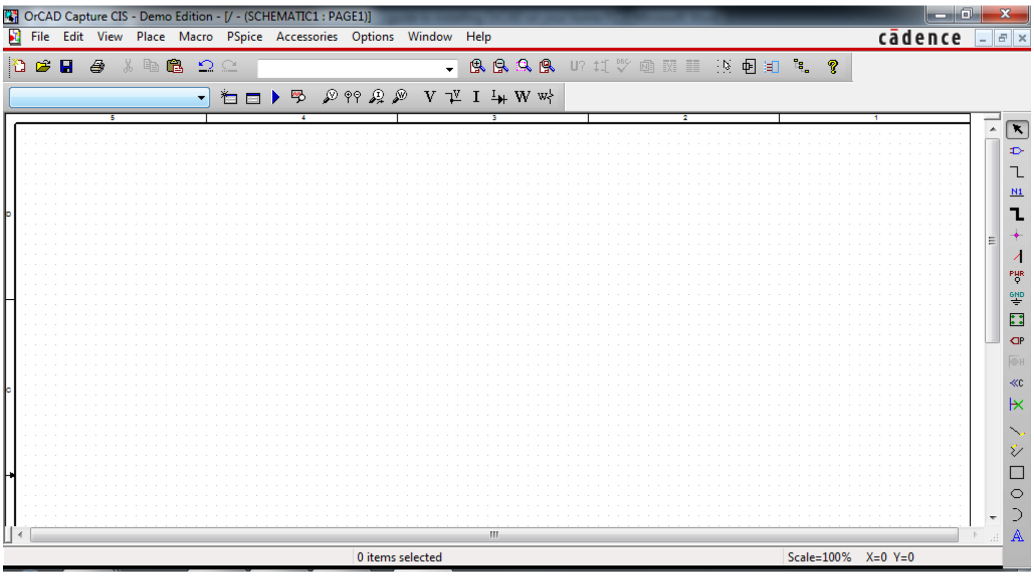
6. When installation is complete, click finish.

7. If you had problems installing the software, complete the practical on one of the computers in a lab on campus.

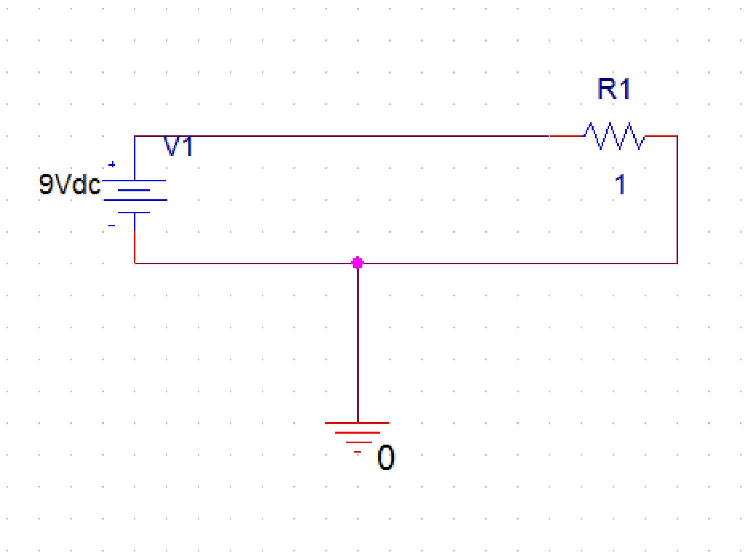
8. After installation, follow the steps below to design your first circuit.

Creating your first OrCAD simulation.

1. Click on Start-> All Programs -> Orcad 16.0 Demo -> OrCAD Capture CIS Demo
2. This will open OrCAD Capture.
3. Select File -> New Project .
4. Give your design a name, Select Analog or mixed A/D, choose a location where you want to save your project and click on OK. Select create a blank project.
5. You should see the circuit schematic in front of you (see below)



6. Press “p” or select “place part” from the tab on the right.
7. Select “Add library” and select all the .olb files(from Amplifier to Transistor). Click open. This allows you to use all the components that you wish to build a circuit with. Also add the .olb files that are in the pspice folder.
8. Select “Place part” again and type in the “Part” field the letter “R”. Select the Resistor R/Analog. Select OK. Double click on the value of the resistor and change its value to 1 ohm
9. Place the part on your schematic.
10. Add a battery to your schematic as well. It has a part name “VDC”. Double click on the battery’s value and change it to 9 V.
11. Add a ground to your schematic. Gnd can be found on the tab to the right. Select the 0/SOURCE ground.
12. Connect the components with wire so that you have a similar circuit to the figure below.



13. You are now ready to simulate your first circuit.
14. Click on PSpice-> New Simulation Profile and give your simulation a name.
15. Select your analysis type to be Time Domain analysis and change the run to time to 3 ms. Click Ok.
16. Place a current marker (found in the menu above the schematic – It has a probe icon with an I inside it). Place it in order to measure the current flowing through the resistor.
17. Click on the Run Pspice button (a blue arrow pointing to the right)
18. A graph should appear indicating that the current flowing through the resistor is 9 A.

You now understand the basics of OrCAD and can now move on to the pre-practical assignment.